X-ray Diffraction from Crystals

Path difference, structure factor, and systematic absences

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Geometry and assumptions

Let the incident and scattered wavevectors be $\mathbf{k}_i$ and $\mathbf{k}_f$, with $|\mathbf{k}_i| = |\mathbf{k}_f| = k = 2\pi/\lambda$. Define the scattering vector

$$\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i.$$

- Elastic scattering: $|\mathbf{k}_i| = |\mathbf{k}_f|$.
- A lattice point is $\mathbf{R}_{mn} = m\mathbf{a}_1 + n\mathbf{a}_2$.
- The atomic structure factor is taken angle-independent: $f_j(\theta) = f_j$.

For a symmetric reflection, the angle between the incident beam and the lattice planes is θ, hence the angle between the incident and diffracted beams is 2θ.

Path difference

For atoms at $\mathbf{r}$ and $\mathbf{r}+\mathbf{R}$, the extra distance travelled by the incident ray is $-\hat{\mathbf{k}}_i \cdot \mathbf{R}$, while that on exit is $+\hat{\mathbf{k}}_f \cdot \mathbf{R}$. Therefore

$$\boxed{\Delta(\mathbf{R}) = (\hat{\mathbf{k}}_f - \hat{\mathbf{k}}_i) \cdot \mathbf{R}}$$

and the phase difference is

$$\boxed{\phi(\mathbf{R}) = \frac{2\pi}{\lambda}\Delta(\mathbf{R}) = \mathbf{q} \cdot \mathbf{R}.}$$

For $\mathbf{R}_{mn} = m\mathbf{a}_1 + n\mathbf{a}_2$,

$$\phi_{mn} = mq \cdot \mathbf{a}_1 + nq \cdot \mathbf{a}_2$$

Constructive interference occurs when ϕ_{mn} is an integer multiple of 2π for every lattice translation.

Laue condition and Bragg form

The intensity is maximized when

$$\mathbf{q}\cdot \mathbf{a}_1 = 2\pi h, \qquad \mathbf{q}\cdot \mathbf{a}_2 = 2\pi k,$$

where $h, k \in \mathbb{Z}$. In three dimensions this is

$$\mathbf{q} = \mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3,$$

with $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$.

$$|\mathbf{G}_{hkl}| = \frac{2\pi}{d_{hkl}}, \qquad |\mathbf{q}| = 2k\sin\theta = \frac{4\pi}{\lambda}\sin\theta.$$

Thus

$$\boxed{2d_{hkl}\sin\theta = n\lambda}$$

which is Bragg's law. The integer n can be absorbed into the Miller indices, so the first-order reciprocal-lattice condition is normally used.

Structure factor

Let the unit cell contain atoms $j = 1, \dots, N$, at fractional coordinates (x_j, y_j, z_j), with $\mathbf{r}_j = x_j\mathbf{a}_1 + y_j\mathbf{a}_2 + z_j\mathbf{a}_3$. The scattered amplitude is

$$A(\mathbf{q}) = \sum_{\mathbf{R}} \sum_j f_j \exp[i\mathbf{q}\cdot(\mathbf{R} + \mathbf{r}_j)].$$

At a reciprocal-lattice point, $\exp(i\mathbf{q}\cdot\mathbf{R}) = 1$, and the unit-cell amplitude is the structure factor

$$\boxed{F_{hkl} = \sum_{j=1}^{N} f_j \exp\{2\pi i(hx_j + ky_j + lz_j)\}}.$$

The diffracted intensity is

$$\boxed{I_{hkl} \propto |F_{hkl}|^2}.$$

If all atoms are identical with constant f, then $F_{hkl} = f\sum_j e^{2\pi i(hx_j + ky_j + lz_j)}$.

Simple cubic (SC)

The conventional SC cell has one lattice point:

$$(0, 0, 0).$$

Therefore

$$F_{hkl} = f, \qquad I_{hkl} \propto f^2.$$

Extinction rule

There are no systematic absences: all integer (hkl) are allowed, apart from the trivial exclusion (000) as a Bragg reflection.

Body-centred cubic (BCC)

Basis positions:

$$(0, 0, 0), \qquad \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right).$$

Hence

$$F_{hkl} = f\left[1 + e^{\pi i(h+k+l)}\right] = f\left[1 + (-1)^{h+k+l}\right].$$

Extinction rule

$h + k + l = 2n$: allowed; $h + k + l = 2n + 1$: absent.

Face-centred cubic (FCC)

Basis positions:

$$(0,0,0),\ (0,\tfrac{1}{2},\tfrac{1}{2}),\ (\tfrac{1}{2},0,\tfrac{1}{2}),\ (\tfrac{1}{2},\tfrac{1}{2},0).$$

Thus

$$F_{hkl} = f\left[1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}\right].$$

Extinction rule

Reflections occur only when h, k, l are all even or all odd. Mixed parity gives $F_{hkl} = 0$.

For allowed reflections, $|F| = 4f$.

Hexagonal close-packed (HCP)

For the primitive hexagonal cell, use the two-point basis

$$(0,0,0), \qquad \left(\tfrac{2}{3},\tfrac{1}{3},\tfrac{1}{2}\right).$$

The basis factor is

$$F_{hkl} = f\left[1 + e^{2\pi i(2h+k)/3} e^{\pi i l}\right].$$

Equivalently,

$$F_{hkl} = f\left[1 + (-1)^{l} e^{2\pi i(2h+k)/3}\right].$$

Extinction rule (ideal HCP basis)

The basis factor vanishes only when l is odd and $2h+k \equiv 0 \pmod 3$. For even l, it is nonzero for every h, k.

Diamond cubic (DC)

Diamond cubic is an FCC lattice plus the basis translation

$$\mathbf{t} = \left(\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right).$$

Its structure factor factors as

$$F_{hkl} = f\, S_{\mathrm{FCC}}(hkl)\left[1 + e^{\frac{i\pi}{2}(h+k+l)}\right],$$

where $S_{\mathrm{FCC}} = 1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}$.

Extinction rule

First require FCC parity: h, k, l all even or all odd. If all even, reflections with $h + k + l \equiv 2 \pmod 4$ are absent. If all odd, the diamond factor is nonzero; hence all-odd reflections are allowed.

Examples: Si allows 111, 220, and 400; 222 is absent.

One-page rule sheet

Lattice	Systematic condition for nonzero F_{hkl}
SC	All hkl allowed.
BCC	$h+k+l$ even.
FCC	h,k,l all even or all odd.
HCP basis	Nonzero except when l is odd and $2h+k \equiv 0 \pmod 3$.
Diamond cubic	FCC condition plus, for all-even indices, $h+k+l \not\equiv 2 \pmod 4$.

Important distinction

These are systematic absences caused by translational symmetry. Additional zeros can arise from the actual atomic form factors, multiple chemical species, disorder, thermal motion, or non-ideal atomic positions.

Conventions and checks

- A different origin changes F_{hkl} by an overall phase, not the intensity $|F_{hkl}|^2$.
- HCP extinction statements depend on the chosen hexagonal basis and on whether one is describing the ideal two-atom basis or a larger conventional cell.
- For a multi-element crystal, replace the common f by the appropriate f_j in the general structure-factor sum.
- The angle-independent-atom assumption isolates geometric interference; real X-ray form factors generally depend on angle through $\sin\theta/\lambda$.

BCC selection rules

Plane	N	Plane	N	Plane	N
100	1	221	9	**411, 330**	**18**
110	**2**	**310**	**10**	331	19
111	3	311	11	**420**	**20**
200	**4**	**222**	**12**	421	21
210	5	320	13	**332**	**22**
211	**6**	**321**	**14**	**422**	**24**
220	**8**	**400**	**16**	**431**	**26**
		410	17	432	29

Allowed reflections shown in orange

For BCC, $h + k + l$ must be even; equivalently, the allowed values of $N = h^2 + k^2 + l^2$ in this range are even.

FCC selection rules

Plane	N	**Plane**	N	**Plane**	N
100	1	221	9	411, 330	18
110	2	310	10	**331**	**19**
111	**3**	**311**	**11**	**420**	**20**
200	**4**	**222**	**12**	421	21
210	5	320	13	332	22
211	6	321	14	**422**	**24**
220	**8**	**400**	**16**	431	26
		410	17	432	29

Allowed reflections shown in orange

For FCC, h, k, l must be all even or all odd; mixed parity is systematically absent.

Diamond cubic selection rules

Plane	N	Plane	N	Plane	N
100	1	221	9	411, 330	18
110	2	310	10	**331**	**19**
111	**3**	**311**	**11**	420	20
200	4	222	12	421	21
210	5	320	13	332	22
211	6	321	14	**422**	**24**
220	**8**	**400**	**16**	431	26
		410	17	432	29

Allowed reflections shown in orange

Diamond cubic first obeys FCC parity; all-even reflections with $h + k + l \equiv 2 \pmod 4$ are additionally absent.

$\sin^2\theta$ ratios for allowed reflections

SC	$1:2:3:4:5:6:8:\ldots$
BCC	$1:2:3:4:5:6:7:\ldots$
FCC	$3:4:8:11:12:\ldots$
DC	$3:8:11:16:\ldots$

How to use the sequence

For a cubic crystal, $\sin^2\theta = (\lambda^2/4a^2)N$, where $N = h^2 + k^2 + l^2$. Each row is shown in its conventional integer scale; the BCC row is the allowed N sequence divided by 2.

Example 1: BCC lattice parameter

Given $\sin^2\theta = 0.120, 0.238, 0.357, 0.475, 0.593, 0.711, 0.830$,

$$\frac{\sin^2\theta_i}{0.120} = 1.000, 1.983, 2.975, 3.958, 4.942, 5.925, 6.917,$$
$$\approx 1 : 2 : 3 : 4 : 5 : 6 : 7.$$

This is the BCC sequence $N = 2, 4, 6, 8, 10, 12, 14$, indexed as

$$(110), (200), (211), (220), (310), (222), (321).$$

Using Cu $K\alpha$ radiation, $\lambda = 1.54\,$Å,

$$a_i = \frac{\lambda\sqrt{N_i}}{2\sqrt{\sin^2\theta_i}} = 3.144, 3.157, 3.157, 3.160, 3.162, 3.163, 3.162\,\text{Å}.$$

Result

The crystal is **BCC**, with $\bar{a} = 3.158\,\text{Å} \approx \boxed{3.16\,\text{Å}}$.

Example 2: diamond cubic

For $\theta_1 = 34.06^\circ$, $\theta_2 = 65.51^\circ$, and Cr $K\alpha$ radiation $(\lambda = 2.29\,\text{Å})$,

$$\sin^2\theta_1 = 0.3137, \qquad \sin^2\theta_2 = 0.8282, \qquad \frac{\sin^2\theta_2}{\sin^2\theta_1} = 2.640 \approx \frac{8}{3}.$$

The sequence is $N = 3\ (111)$, $N = 8\ (220)$. FCC would also allow $N = 4\ (200)$; its absence identifies **diamond cubic.**

$$a_{111} = \frac{2.29\sqrt{3}}{2\sin 34.06^\circ} = 3.541\,\text{Å}, \qquad a_{220} = \frac{2.29\sqrt{8}}{2\sin 65.51^\circ} = 3.559\,\text{Å}.$$

The next allowed DC line is $N = 11\ (311)$. With $a \approx 3.55\,\text{Å}$,

$$\sin\theta_3 = \frac{\lambda\sqrt{11}}{2a} \approx 1.07 > 1.$$

Result

$a \approx \boxed{3.55\,\text{Å}}$. Since $\sin\theta_3 > 1$, the (311) line and every higher line are impossible.

Example 3: Cu $K\alpha$ energy

For $\lambda = 1.5406\,$Å $= 1.5406 \times 10^{-10}$ m,

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^{8})}{1.5406 \times 10^{-10}} = 1.2903 \times 10^{-15}\ \text{J}.$$

$$\frac{E}{e} = 8.054 \times 10^{3}\ \text{eV} = 8.054\,\text{keV}, \qquad V_{\text{equiv}} = \frac{E}{e} \approx \boxed{8.05\,\text{kV}}.$$

The Cu K-edge is $8.98\,$keV, so the ideal threshold for ejecting a K-shell electron is

$$eV_{\text{min}} = 8.98\,\text{keV} \quad \Rightarrow \quad \boxed{V_{\text{min}} = 8.98\,\text{kV}}.$$

Why the K-edge voltage is higher

Creating the vacancy costs the full K-shell binding energy. A $K\alpha$ photon carries only the difference between the K- and L-shell energies, about $8.98 - 8.054 = 0.926\,$keV less. Practical tubes therefore operate above the threshold.

Thank You

Questions?